MUSKAN CHAUHAN

Unity Game Developer / XR Developer

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Portfolio | GitHub

PROFESSIONAL SUMMARY

Dedicated Unity Game Developer bridging commercial mobile gaming and enterprise XR simulations. Experienced in managing end-to-end application deployment, from architecting scalable core gameplay mechanics to optimizing cross-platform performance for mobile and VR hardware. Highly adaptable and quick to master new development stacks, blending creative problem-solving with technical rigor to translate complex requirements into engaging, high-performance products.

TECHNICAL SKILLS

Unity (C#) | XR Development | Digital Twins | Rest APIs and Data Handling | Gameplay Programming | Game Design Patterns | Object Pooling | Multiplayer Architecture (Photon PUN) | Mobile Optimization | Shader Graph | Particle Systems | Firebase Analytics | VR Simulations

WORK EXPERIENCE

Unity Developer | Battlebucks

Sep 2025 - Present | Gurugram, Haryana, India

Built and shipped games for Battlebucks' multi-game Android platform. Each game is its own module in a shared Unity hub project and is loaded into the native Android app on demand through Addressables. A shared in-house SDK handles the game lifecycle, UI, scoring and difficulty config sent from the server.

- Implemented Firebase Analytics SDK to track custom gameplay events and user behavior, translating data into actionable insights for mechanic balancing and player retention.
- Utilized Firebase telemetry to monitor player progression and in-app events, creating pipelines to optimize ad placements and the overall user experience.

Hungry Shadow — 2D Action Brawler (Android)

Unity (C#) · Spine 2D · Addressables · Particle Systems · Custom Shaders

- **Gameplay:** Built a responsive combat system with a state machine covering standing, crouch, air and weapon combos. It takes tap and swipe input through an input buffer that expires old inputs, so fast repeated taps don't fire late or in the wrong direction.
- **Enemies:** Designed a shared enemy AI base and extended it into 8 enemy types with distinct attacks (melee, ranged, shields, throws). It includes knockback, knock-up and air-juggle physics, plus a dismemberment system whose weapon drops the player can catch and throw.
- **Game feel:** Added hit feedback with pooled VFX, custom shaders, camera-shake patterns tied to combo hits, and Spine animation blending synced to the knockback movement.
- **Performance:** Kept gameplay at a steady 80–90 FPS on mid-range Android phones by pooling objects and VFX, preloading audio and shader variants, and removing per-frame memory allocations.

- **Architecture:** Kept the game manager to lifecycle control only and put each feature (spawning, scoring, stats) in its own component. That kept the SDK integration to one small surface, and it meant difficulty could come from server-side config without a new build. Chose a state machine for combat so each new combo is its own state rather than another branch in one growing *if* chain.

Shadow Jump — 2D Arcade Platformer (Android)

Unity (C#) · Feel (MMFeedbacks) · Custom URP Shader · Animator

- **Game-feel overhaul:** Built a feedback system on the Feel asset with 15+ custom feedbacks for jump, double-jump, landing, dash, trail and death. Effects included squash-and-stretch, particle and sprite bursts, and landing effects that scale with how hard the player lands.
- **Custom shader:** Wrote a URP sprite shader that leans the character from the feet up, driven by spring motion. It uses MaterialPropertyBlocks, so no extra materials are created and sprites still batch together.
- **Architecture:** Replaced the single player animator script with an event-driven design. The Animator handles movement blending, gameplay events trigger the visual and audio feedback, and a shared component controls the sprite's shape changes. Gameplay code never references effects directly, so new effects can be added or retuned without risking gameplay bugs.
- **Gameplay and polish:** Reworked the double jump, replaced physics colliders with a custom capsule check for fairer sawblade hits, and fixed the sawblade movement and spawning logic. Also fixed timer, collider, animation and UI bugs, and added a fixed-aspect-ratio camera so the game looks the same on every screen shape.

Jr. XR Developer | Simulanis

January 2023 - August 2025 | Noida, Uttar Pradesh

- Engineered gamified training modules and complex VR simulators across multiple platforms (Mobile, Oculus, WebGL) for diverse industries.
- Architected a proprietary cross-platform Metaverse, streamlining the internal pipeline to achieve a 60% reduction in development time.
- Developed immersive VR projects featuring guided and unguided training modalities, integrating performance analytics for actionable insights into user progress.
- Delivered complex hardware-integrated experiences, specifically a forklift simulator featuring real-time data tracking for user performance.
- Optimized application performance and user experience (UX) by rigorously applying VR best practices.
- Researched and integrated emerging XR technologies and plugins from leading platforms, including Meta and Pico.
- Managed version control and collaborative development lifecycles utilizing Git and GitHub to ensure code integrity.

PERSONAL PROJECTS

Grim: Multiplayer 2D Arena Shooter (Android) *(Collaborative Project)* ▶ Get it on Google Play

Roles: Gameplay Programming, Game Feel & VFX, UI · Unity 6 (C#) · Photon Fusion · Unity Gaming Services

- A fast-paced online arena shooter with short, deadly rounds. Weapons are scattered across the map and players race to grab them. Built with Photon Fusion (host/client) networking and Unity Gaming Services for accounts, cloud saves, economy, leaderboards, friends and voice chat.
- **Weapons and gameplay:** Built weapons that can be picked up and thrown (Pistol, Assault Rifle, Shotgun, Sniper and Grenade) on a shared weapon base class. Features include recoil, floor pickups, throwing and aim assist, with the server deciding every grab so two players can never take the same weapon.

- **Game feel and VFX:** Built the combat feedback with Feel: hit-stop on kills, camera punches, squash-and-stretch, and squash-and-spring touch feedback on the combat buttons. VFX come from a pooled, capped system so busy 4-player fights stay smooth on phones, and all tuning lives in one ScriptableObject asset.
- **UI and meta systems:** Designed and built the full menu flow for a mobile competitive game: home hub, hero gallery and mastery, daily rewards, chests, missions, and friends and chat. Navigation is consistent across screens and the controls are icon-only. Each screen talks to the backend only through an interface, so backend services can be swapped without changing the UI.
- **Collaboration:** Co-developed with another developer, with each of us owning separate modules (gameplay, feel and UI versus networking and backend). We integrated through Git with feature branches and code reviews, against a shared game design document and milestone plan.

Neon Bricks: Hyper-Casual Brick Breaker ▶ Closed Beta on Google Play

Roles: Programming, Game Design, Product Planning

- Developed a multi-platform (Android, WebGL) game in Unity 6, architecting a scalable codebase utilizing State Machine, Observer, and Strategy design patterns based on SOLID principles.
- Designed and implemented a targeted Firebase Analytics event plan to monitor player retention and level difficulty, utilizing clean telemetry for actionable live-ops insights.
- Integrated Google AdMob (rewarded and interstitial) behind a custom service interface and successfully published the optimized build to the Google Play Store.
- Authored 50 data-driven levels using ScriptableObjects, optimizing memory and rendering performance through robust object pooling and MaterialPropertyBlocks.
- Available on Itch and PlayStore.

Multiplayer Mobile Game (3D)

Roles: Programming, Game Design

- Developed a 3D online FPS mobile game utilizing Unity (C#) and Photon PUN for real-time multiplayer networking.
- Architected scalable core gameplay logic by implementing the State Machine design pattern.

EDUCATION

Bachelor of Technology (B.Tech)

DCRUST, Haryana

January 2018 - December 2022

CERTIFICATIONS & COURSES

- Core Java - HP
- Java Specialization (Coursera)
- JDBC Programming (Udemy)
- Multiplayer Unity Games - Photon PUN2 (Udemy)